

Anti-seizure Pharmacology

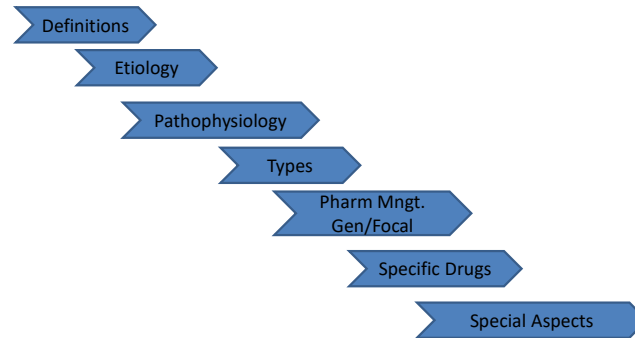


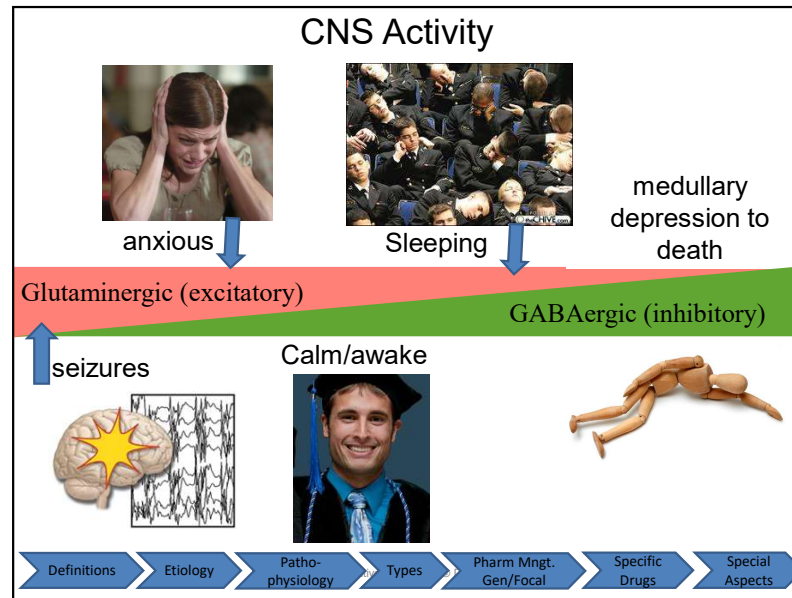
Chapter 24, Basic and Clinical Pharmacology, 11th or 12th. edition. Katzung.
Material in Harrison's Chapter 363 will help in your understanding of this subject.

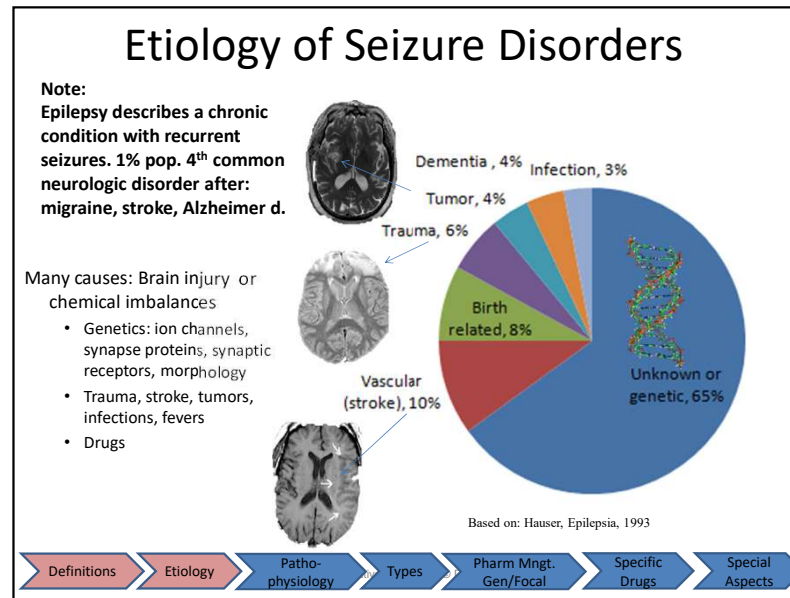
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Today's Topics







Note:

Epilepsy describes a chronic condition with recurrent seizures. (e.g two or more unprovoked seizures)

Neonates <1 month

Hypoxia, trauma, infection (bacterial meningitis). Metabolic: uremia (excess urea in blood), electrolyte disorders, metabolic (hyponatremia, porphyria, hypoglycemia, hyperglycemia, hypomagnesemia, phenylketonuria). Cardiac arrest (imbalance of oxygen, CO poisoning), drug withdrawal, genetic disorders, developmental.

Infants (>1 month and >12 months)

Febrile (6 months to 5 years) benign condition, genetic, infection, developmental, trauma idiopathic

Adolescents (12-18 years)

Trauma, genetic, infection, brain tumor, illicit drug use, idiopathic

Young Adults (18-35 years)

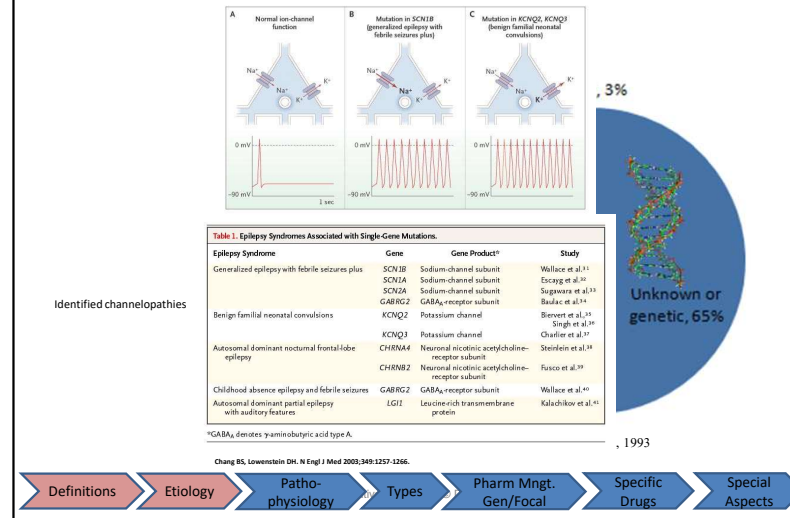
Trauma, alcohol withdrawal, illicit drug use, brain tumor, idiopathic

Older Adults (>35years)

Cerebrovascular disease, brain tumor, alcohol withdrawal, Metabolic (uremia, hepatic failure, thyrotoxic storm, electrolyte abnormalities, hypoglycemia), Alzheimer's disease, idiopathic

From Harrison's chapter 445

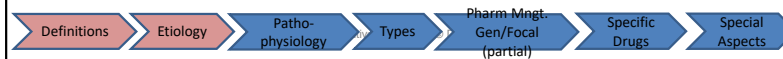
Etiology of Seizure Disorders



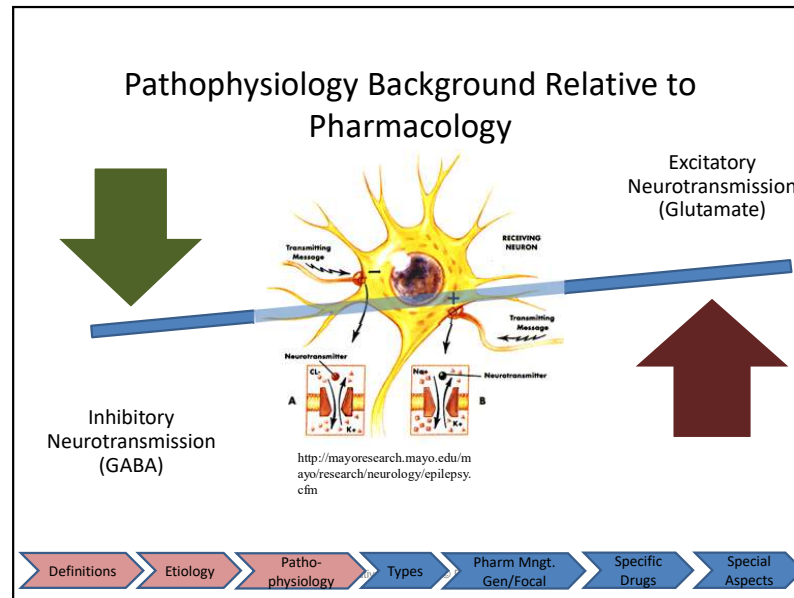
Katzung's text: A wide diversity of genes may be affected, including (1) those encoding voltage-gated ion channels such as voltage-gated sodium channels and synaptic receptors such as GABA_A receptors, (2) components of the neurotransmitter release machinery including syntaxin binding protein (STXBP1), (3) neural adhesion molecules such as protocadherin 19 (PCDH19), and (4) proteins involved in synapse development such as leucine-rich glioma inactivated-1 (LGI1).

Drugs that ↓ seizure threshold

- Alkylating agents(e.g. busulfan, chlorambucil)
- Antimalarials.....(chloroquine, mefloquine)
- Antimicrobials/ antibiotics .(b-lactams, quinolones, acyclovir, isoniazid, ganciclovir)
- Anesthetic and analgesic.....(meperidine, tramadol, local anesthetics)
- Dietary supplements.....(ephedra (ma huang), Gingko)
- Immunomodulatory drugs ...(cyclosporine, tacrolimus, interferons)
- Some Antidepressants, antipsychotics
- lithium
- Theophylline
- Sedative hypnotic withdrawal
- Drugs of abuse..... (cocaine, phencyclidine, amphetamine, methylphenidate)
- Flumazenil

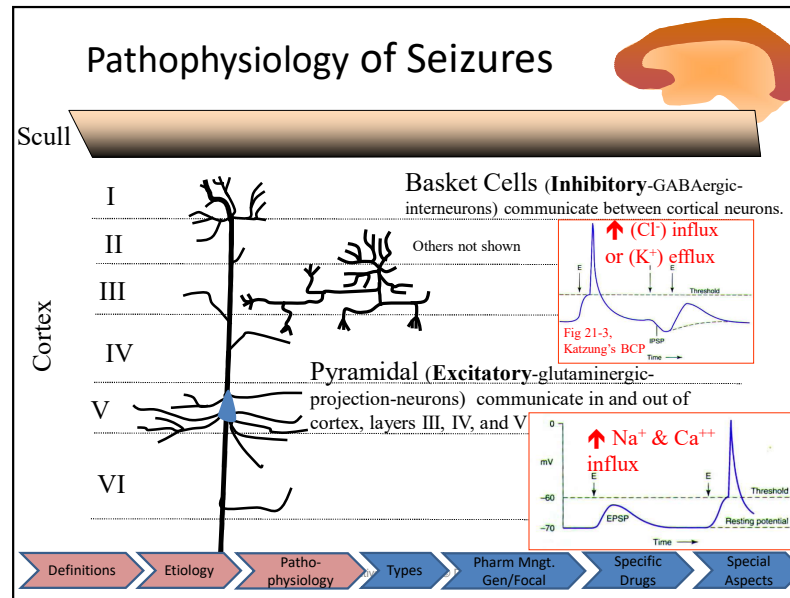


List drugs associated with decreasing seizure threshold



A seizure occurs when the message delivery system becomes unbalanced. Under normal circumstances, the neurotransmitter GABA does its part to make sure the system stays in synch by triggering signals in the form of charged particles (A). It causes a large concentration of negatively charged chloride particles (Cl^-) to enter the receiving neuron. This tells the neuron to not pass on the message. When there is not enough GABA a person can have a seizure because receiving neurons can be flooded with signals that say "pass on this message." The "go" messages are triggered by a different type of neurotransmitter that promotes message transfer (B). The charged signals they set off are positive. This time, more positively charged sodium particles (Na^+) enter the neuron, which tells the receiving neuron to pass on the message.

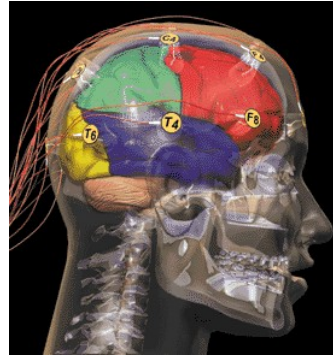
Describe the pathophysiology of Focal (partial) and generalized seizure and how this relates to GABAergic and Glutaminergic neuronal activity

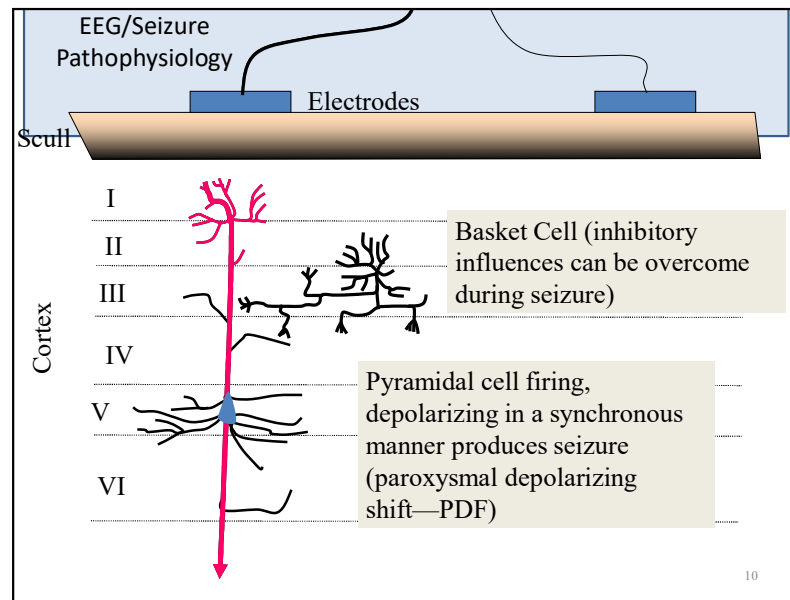


Describe the pathophysiology of Focal (partial) and generalized seizure and how this relates to GABAergic and Glutaminergic neuronal activity

Understand the molecular role of GABA mimetic drugs and drugs that decrease sodium and calcium current in reducing seizure activity.

Electroencephalogram (EEG): Detection of brain wave activity





Describe the pathophysiology of Focal (partial) and generalized seizure and how this relates to GABAergic and Glutaminergic neuronal activity

Recognize the EKG patterns, especially with petit mal (Absence seizures)

Understand the molecular role of GABA mimetic drugs and drugs that decrease sodium and calcium current in reducing seizure activity.

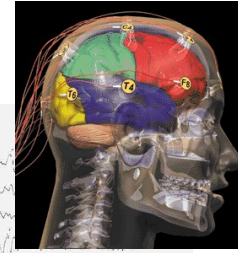
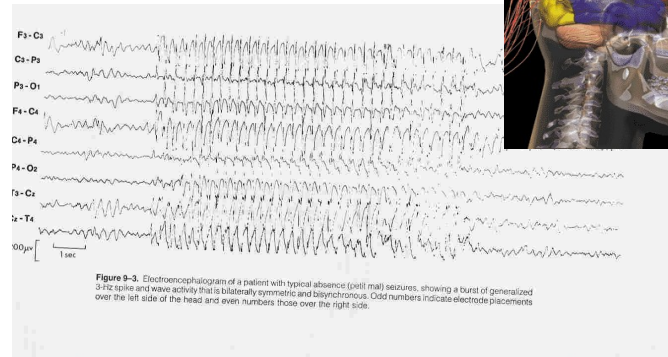
From Harrison's 445

MECHANISMS OF SEIZURE INITIATION AND PROPAGATION

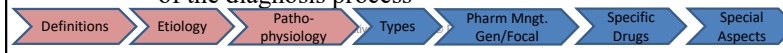
Focal seizure activity can begin in a very discrete region of cortex and then slowly invade the surrounding regions. The hallmark of an established seizure is typically an electrographic “spike” due to intense near-simultaneous firing of a large number of local excitatory neurons, resulting in an apparent

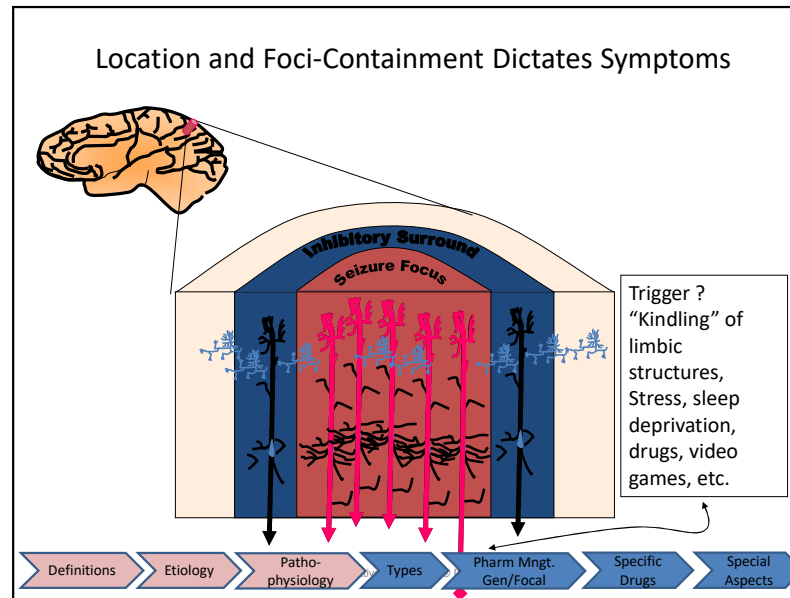
hypersynchronization of the excitatory bursts across a relatively large cortical region. The bursting activity in individual neurons (the “paroxysmal depolarization shift”) is caused by a relatively long-lasting depolarization of the neuronal membrane due to influx of extracellular calcium (Ca^{2+}), which leads to the opening of voltage-dependent sodium (Na^+) channels, influx of Na^+ , and generation of repetitive action potentials. This is followed by a hyperpolarizing afterpotential mediated by γ -aminobutyric acid (GABA) receptors or potassium (K^+) channels, depending on the cell type. The synchronized bursts from a sufficient number of neurons result in a so-called spike discharge on the EEG.

EEG (seizure)



History, physical exam, EEG, and MRI are all part of the diagnosis process





Target objective for slide: Background for following objectives

Be able to identify classic symptoms of each type of seizure

Understand underlying molecular mechanism of GABA mimetic drugs and drugs that reduce sodium and calcium current

Know which drugs are used for the specific seizure types

From Harrison's 445

The spreading seizure wavefront is slowed and ultimately halted by intact hyperpolarization and a "surround" inhibition created by feedforward activation of inhibitory neurons. With sufficient activation, there is a recruitment of surrounding neurons via a number of synaptic and nonsynaptic mechanisms, including: (1) an increase in extracellular K^+ , which blunts hyperpolarization and depolarizes neighboring neurons; (2) accumulation of Ca^{2+} in presynaptic terminals, leading to enhanced neurotransmitter release; (3) depolarization-induced activation of the *N*-methyl-d-aspartate (NMDA) subtype of the excitatory

amino acid receptor, which causes additional Ca^{2+} influx and neuronal activation; and (4) ephaptic interactions related to changes in tissue osmolarity and cell swelling. The recruitment of a sufficient number of neurons leads to the propagation of excitatory currents into contiguous areas via local cortical connections and to more distant areas via long commissural pathways such as the corpus callosum.

Many factors control neuronal excitability, and thus there are many potential mechanisms for altering a neuron's propensity to have bursting activity. Mechanisms *intrinsic* to the neuron include changes in the conductance of ion channels, response characteristics of membrane receptors, cytoplasmic buffering, second-messenger systems, and protein expression as determined by gene transcription, translation, and posttranslational modification. Mechanisms *extrinsic* to the neuron include changes in the amount or type of neurotransmitters present at the synapse, modulation of receptors by extracellular ions and other molecules, and temporal and spatial properties of synaptic and nonsynaptic input. Nonneural cells, such as astrocytes and oligodendrocytes, have an important role in many of these mechanisms as well.

Seizure Phenotypes

- Symptoms depend on region effected (E.g..)
 - » Motor, sensory or autonomic
 - » Consciousness (sometimes aura precedes)
 - » Awareness
 - » Memory
 - » Psychic symptoms

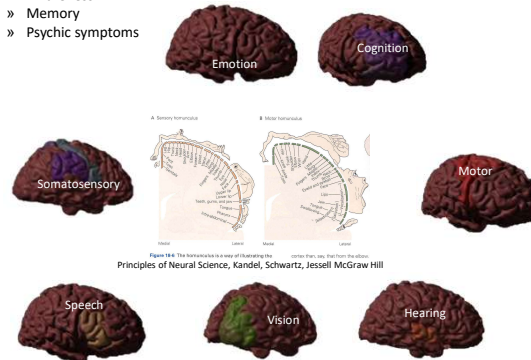
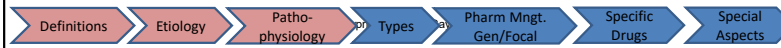
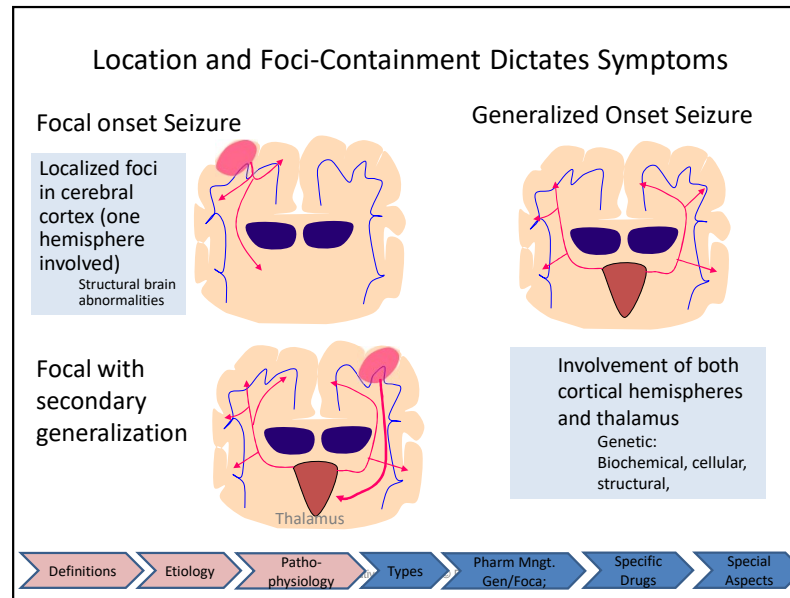


Figure 14.6 The homoculata in a view of the homoculata
Principles of Neural Science, Kandel, Schwartz, Jessell McGraw Hill



Target objective for slide:

Be able to identify classic symptoms of each type of seizure



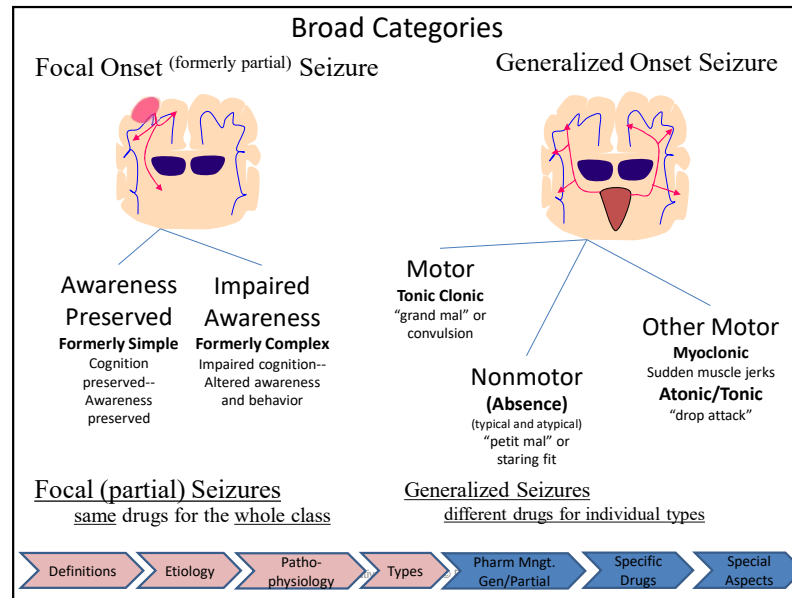
Note: partial seizures occur in discrete regions—consciousness is preserved.

Target objective for slide: Background for following objectives

Be able to identify classic symptoms of each type of seizure

Understand underlying molecular mechanism of GABA mimetic drugs and drugs that reduce sodium and calcium current

Know which drugs are used for the specific seizure types



Focal (partial) Seizures:
 Located on one side of hemispheres.
 Symptoms depend on location (motor, somatosensory, etc.)
 Focal (partial) Seizures account for 60% of all epilepsies
 Etiology of Part seizures: Tumor, developmental malformation, damage due to trauma or stroke

Generalized seizures account for 40% of all epilepsies
 Etiology is usually genetic (perhaps from cellular or biochemical, or structural abnormalities)
 Most common generalized is juvenile myoclonic epilepsy (10% of all epileptic syndromes)
 Arise from the reciprocal firing of the thalamus and cerebral cortex. Central to thalamic oscillations (3 per second) spike and wave are T type calcium channels

Atonic Seizures:
 Mostly children, also adults
 Sudden loss of postural tone (1-2 s); sudden fall
 Wear helmets to prevent head injured

From Harrison's Chapter 445:
Myoclonic jerking:
 Common, seen in a variety of seizures
 Shocklike muscle contractions
 also exists as major seizure type
 Associated with metabolic disorders, degenerative CNS diseases, or anoxic brain injury

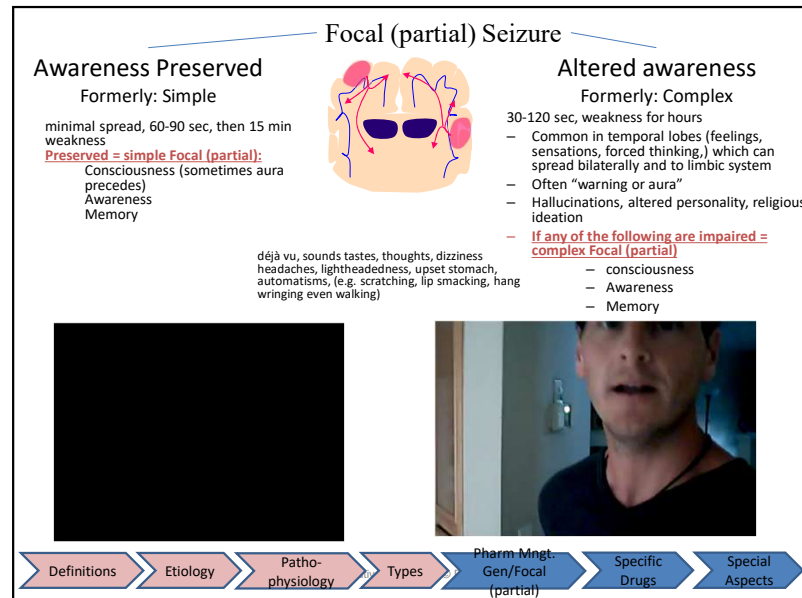
http://www.ninds.nih.gov/disorders/myoclonus/detail_myoclonus.htm
<http://www.ninds.nih.gov/index.htm>

Although rare cases of myoclonus are caused by an injury to the peripheral nerves (defined as the nerves outside the brain and spinal cord, or the central nervous system), most myoclonus is caused by a disturbance of the central nervous system. Studies suggest that several locations in the brain are involved in myoclonus. One such location, for example, is in the brain stem close to structures that are responsible for the startle response, an automatic reaction to an unexpected stimulus involving rapid muscle contraction.

The specific mechanisms underlying myoclonus are not yet fully understood. Scientists believe that some types of stimulus-sensitive myoclonus may involve overexcitability of the parts of the brain that control movement. These parts are interconnected in a series of feedback loops called motor pathways. These pathways facilitate and modulate communication between the brain and muscles. Key elements of this communication are chemicals known as neurotransmitters, which carry messages from one nerve cell, or neuron, to another. Neurotransmitters are released by neurons and attach themselves to receptors on parts of neighboring cells. Some neurotransmitters may make the receiving cell more sensitive, while others tend to make the receiving cell less sensitive. Laboratory studies suggest that an imbalance between these chemicals may underlie myoclonus.

Some researchers speculate that abnormalities or deficiencies in the receptors for certain neurotransmitters may contribute to some forms of myoclonus. Receptors that appear to be related to myoclonus include those for two important inhibitory neurotransmitters: serotonin and gamma-aminobutyric acid (GABA). Other receptors with links to myoclonus include those for opiates and glycine, the latter an inhibitory neurotransmitter that is important for the control of motor and sensory functions in the spinal cord. More research is needed to determine how these receptor abnormalities cause or contribute to myoclonus. Lennox-Gastaut syndrome is a severe form of childhood epilepsy characterized by cognitive impairments and multiple types of seizures including tonic-clonic, tonic, atonic, and myoclonic, and atypical absence seizures.

Target objective for slide: Background for following objectives
 Be able to identify classic symptoms of each type of seizure
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Focal (partial) Seizures:
Symptoms depend on location (motor, somatosensory, etc.)
Focal (partial) Seizures account for 60% of all epilepsies
Etiology of Part seizures: Tumor, developmental malformation, damage due to trauma or stroke

From Harrison's Chapter 445

FOCAL SEIZURES

Focal seizures arise from a neuronal network either discretely localized within one cerebral hemisphere or more broadly distributed but still within the hemisphere. With the new classification system, the subcategories of “simple focal seizures” and “complex focal seizures” have been eliminated. Instead, depending on the presence of cognitive impairment, they can be described as focal seizures with or without dyscognitive features. Focal seizures can also evolve into generalized seizures. In the past this was referred to as focal seizures with secondary generalization, but the new system relies on specific descriptions of the type of generalized seizures that evolve from the focal seizure.

The routine interictal (i.e., between seizures) electroencephalogram (EEG) in patients with focal seizures is often normal or may show brief discharges termed epileptiform spikes, or sharp waves. Because focal seizures can arise from the medial temporal lobe or inferior frontal lobe (i.e., regions distant from the scalp), the EEG recorded during the seizure may be nonlocalizing. However, the seizure focus is often detected using sphenoidal or surgically placed intracranial electrodes.

Focal Seizures Without Dyscognitive Features

Focal seizures can cause motor, sensory, autonomic, or psychic symptoms without impairment of cognition. For example, a patient having a focal motor seizure arising from the right primary motor cortex near the area controlling hand movement will note the onset of involuntary movements of the contralateral, left hand. These movements are typically clonic (i.e., repetitive, flexion/extension movements) at a frequency of ~2-3 Hz; pure tonic posturing may be seen as well. Since the cortical region controlling hand movement is immediately adjacent to the region for facial expression, the seizure may also cause abnormal movements of the face synchronous with the movements of the hand. The EEG recorded with scalp electrodes during the seizure (i.e., an ictal EEG) may show abnormal discharges in a very limited region over the appropriate area of cerebral cortex if the seizure focus involves the cerebral convexity. Seizure activity occurring within deeper brain structures is sometimes not detected by the standard EEG, however, and may require intracranial electrodes for its detection.

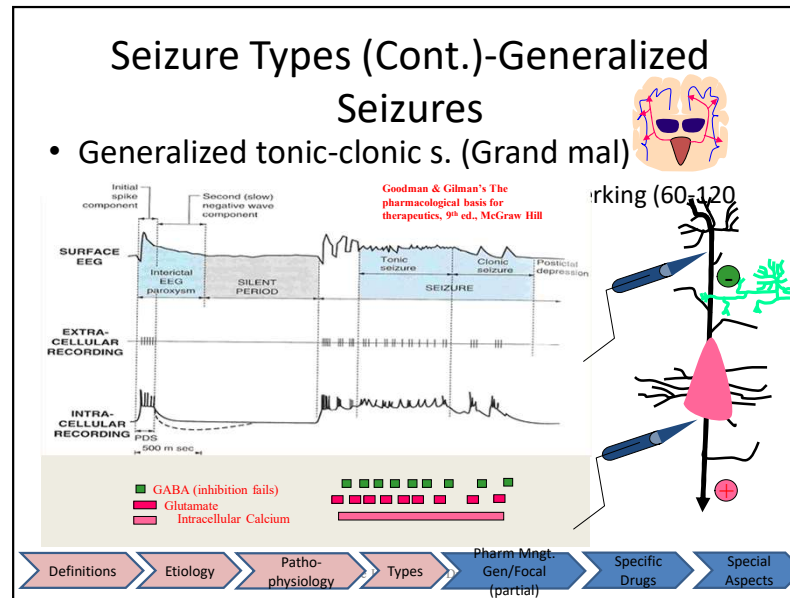
Three additional features of focal motor seizures are worth noting. First, in some patients, the abnormal motor movements may begin in a very restricted region such as the fingers and gradually progress (over seconds to minutes) to include a larger portion of the extremity. This phenomenon, described by Hughlings Jackson and known as a “Jacksonian march,” represents the spread of seizure activity over a progressively larger region of motor cortex. Second, patients may experience a localized paresis (Todd's paralysis) for minutes to many hours in the involved region following the seizure. Third, in rare instances, the seizure may continue for hours or days. This condition, termed *epilepsia partialis continua*, is often refractory to medical therapy.

Focal seizures may also manifest as changes in somatic sensation (e.g., paresthesias), vision (flashing lights or formed hallucinations), equilibrium (sensation of falling or vertigo), or autonomic function (flushing, sweating, piloerection). Focal seizures arising from the temporal or frontal cortex may also cause alterations in hearing, olfaction, or higher cortical function (psychic symptoms). This includes the sensation of unusual, intense odors (e.g., burning rubber or kerosene) or sounds (crude or highly complex sounds), or an epigastric sensation that rises from the stomach or chest to the head. Some patients describe odd, internal feelings such as fear, a sense of impending change, detachment, depersonalization, déjà vu, or illusions that objects are growing smaller (micropsia) or larger (macropsia). These subjective, “internal” events that are not directly observable by someone else are referred to as auras.

Focal Seizures with Dyscognitive Features

Focal seizures may also be accompanied by a transient impairment of the patient's ability to maintain normal contact with the environment. The patient is unable to respond appropriately to visual or verbal commands during the seizure and has impaired recollection or awareness of the ictal phase. The seizures frequently begin with an aura (i.e., a focal seizure without cognitive disturbance) that is stereotypic for the patient. The start of the ictal phase is often a sudden behavioral arrest or motionless stare, which marks the onset of the period of impaired awareness. The behavioral arrest is usually accompanied by automatisms, which are involuntary, automatic behaviors that have a wide range of manifestations. Automatisms may consist of very basic behaviors such as chewing, lip smacking, swallowing, or “picking” movements of the hands, or more elaborate behaviors such as a display of emotion or running. The patient is typically confused following the seizure, and the transition to full recovery of consciousness may range from seconds up to an hour. Examination immediately following the seizure may show an anterograde amnesia or, in cases involving the dominant hemisphere, a postictal aphasia.

The range of potential clinical behaviors linked to focal seizures is so broad that extreme caution is advised before concluding that stereotypic episodes of bizarre or atypical behavior are not due to seizure activity. In such cases additional, detailed EEG studies may be helpful.



From Harrison's Chapter 445

Generalized, Tonic-Clonic Seizures

Generalized-onset tonic-clonic seizures are the main seizure type in ~10% of all persons with epilepsy. They are also the most common seizure type resulting from metabolic derangements and are therefore frequently encountered in many different clinical settings. The seizure usually begins abruptly without warning, although some patients describe vague premonitory symptoms in the hours leading up to the seizure. This prodrome is distinct from the stereotypic auras associated with focal seizures that generalize. The initial phase of the seizure is usually tonic contraction of muscles throughout the body, accounting for a number of the classic features of the event. Tonic contraction of the muscles of expiration and the larynx at the onset will produce a loud moan or “ictal cry.” Respirations are impaired, secretions pool in the oropharynx, and cyanosis develops. Contraction of the jaw muscles may cause biting of the tongue. A marked enhancement of sympathetic tone leads to increases in heart rate, blood pressure, and pupillary size. After 10–20 s, the tonic phase of the seizure typically evolves into the clonic phase, produced by the superimposition of periods of muscle relaxation on the tonic muscle contraction. The periods of relaxation

progressively increase until the end of the ictal phase, which usually lasts no more than 1 min. The postictal phase is characterized by unresponsiveness, muscular flaccidity, and excessive salivation that can cause stridorous breathing and partial airway obstruction. Bladder or bowel incontinence may occur at this point. Patients gradually regain consciousness over minutes to hours, and during this transition, there is typically a period of postictal confusion. Patients subsequently complain of headache, fatigue, and muscle ache that can last for many hours. The duration of impaired consciousness in the postictal phase can be extremely long (i.e., many hours) in patients with prolonged seizures or underlying central nervous system (CNS) diseases such as alcoholic cerebral atrophy.

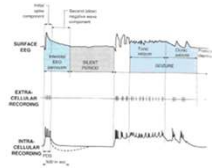
The EEG during the tonic phase of the seizure shows a progressive increase in generalized low-voltage fast activity, followed by generalized high-amplitude, polyspike discharges. In the clonic phase, the high-amplitude activity is typically interrupted by slow waves to create a spike-and-wave pattern. The postictal EEG shows diffuse slowing that gradually recovers as the patient awakens.

There are a number of variants of the generalized tonic-clonic seizure, including pure tonic and pure clonic seizures. Brief tonic seizures lasting only a few seconds are especially noteworthy since they are usually associated with specific epileptic syndromes having mixed seizure phenotypes, such as the Lennox-Gastaut syndrome (discussed below).

Seizure Types (Cont.)-Generalized Seizures

- Generalized tonic-clonic s. (*Grand mal*)

- Symptoms



Goodman & Gilman's The pharmacological basis for therapeutics, 9th ed., McGraw Hill



Target objective for slide:

Be able to identify classic symptoms of each type of seizure

Seizure Types (Cont.)-Generalized Seizures

Absence (*Petit mal*)

- Genetic transmission
- Begin in childhood (y 4-8)
 - Main seizure type in children (15-20%)
 - 60-70% have spontaneous remission during adolescence
- sudden onset, abrupt end (<10 sec) up to 100 times/d
- Characteristic EEG 2-3 Hz spike-and-wave
 - Produced by thalamic T type Calcium channels
 - Therapy involves inhibiting these T-type Ca channels
- Note: atypical absence associated with more diffuse issues, structural abnormalities, and mental retardation. See treatment differences



Target objective for slide:

Be able to identify classic symptoms of each type of seizure

From Harrison's chapter 363—note about atypical absence seizures. “Atypical absence seizures have features that deviate both clinically and electrophysiologically from typical absence seizures. For example, the lapse of consciousness is usually of longer duration and less abrupt in onset and cessation, and the seizure is accompanied by more obvious motor signs that may include focal or lateralizing features. The EEG shows a generalized, slow spike-and-wave pattern with a frequency of 2.5/s, as well as other abnormal activity. Atypical absence seizures are usually associated with diffuse or multifocal structural abnormalities of the brain and therefore may accompany other signs of neurologic dysfunction such as mental retardation. Furthermore, the seizures are less responsive to anticonvulsants compared to typical absence seizures.”

From Harrison's chapter 445

Typical Absence Seizures

Typical absence seizures are characterized by sudden, brief lapses of consciousness without loss of postural control. The seizure typically lasts for only seconds, consciousness returns as suddenly as it was lost, and there is no postictal confusion. Although the brief loss of consciousness may be clinically inapparent or the sole manifestation of the seizure discharge, absence seizures are usually accompanied by subtle, bilateral motor signs such as rapid blinking of the eyelids, chewing movements, or small-amplitude, clonic movements of the hands.

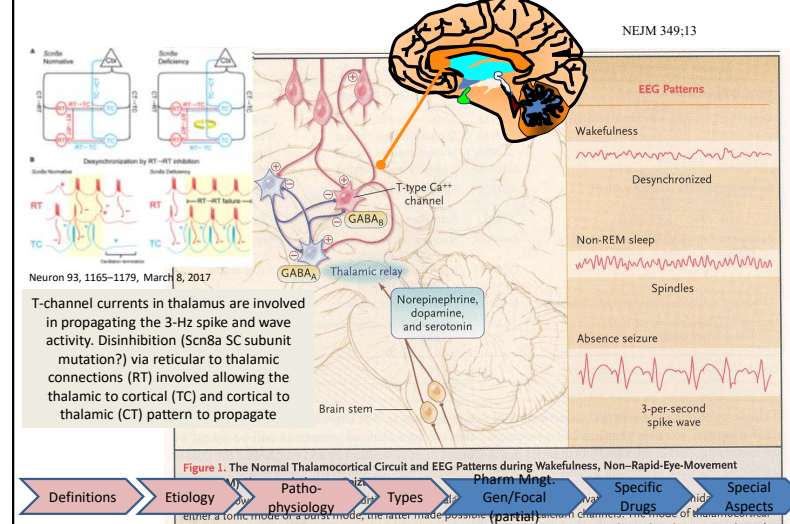
Typical absence seizures are associated with a group of genetically determined epilepsies with onset usually in childhood (ages 4–8 years) or early adolescence and are the main seizure type in 15–20% of children with epilepsy. The seizures can occur hundreds of times per day, but the child may be unaware of or unable to convey their existence. Because the clinical signs of the seizures are subtle, especially to parents who may not have had previous experience with seizures, it is not surprising that the first clue to absence epilepsy is often unexplained “daydreaming” and a decline in school performance recognized by a teacher.

The electrophysiologic hallmark of typical absence seizures is a generalized, symmetric, 3-Hz spike-and-wave discharge that begins and ends suddenly, superimposed on a normal EEG background. Periods of spike-and-wave discharges lasting more than a few seconds usually correlate with clinical signs, but the EEG often shows many more brief bursts of abnormal cortical activity than were suspected clinically. Hyperventilation tends to provoke these electrographic discharges and even the seizures themselves and is routinely used when recording the EEG.

Atypical Absence Seizures

Atypical absence seizures have features that deviate both clinically and electrophysiologically from typical absence seizures. For example, the lapse of consciousness is usually of longer duration and less abrupt in onset and cessation, and the seizure is accompanied by more obvious motor signs that may include focal or lateralizing features. The EEG shows a generalized, slow spike-and-wave pattern with a frequency of ≤ 2.5 per second, as well as other abnormal activity. Atypical absence seizures are usually associated with diffuse or multifocal structural abnormalities of the brain and therefore may accompany other signs of neurologic dysfunction such as mental retardation. Furthermore, the seizures are less responsive to anticonvulsants compared to typical absence seizures.

T-type calcium channel and “absence” seizures



Target objective for slide:

Be able to identify classic symptoms of each type of seizure

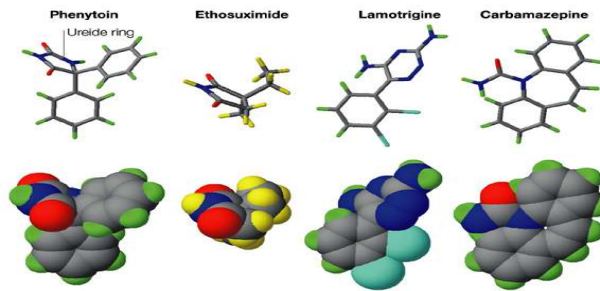
Know which drugs are used for the specific seizure types (illustrates the background behind using drugs like ethosuximide). Also, identifying the 3 hz spike and wave on an exam is important.

3 Hz spike and wave pattern represents oscillations in excitatory communication between the thalamus and the neocortex. Underlying this communication are the low threshold T-type calcium channels that are activated by more negative membrane potentials than other channels (low threshold). In the thalamus these t-type VGCC are prevalent.

Pharmacologic Management of Epilepsy

Drug Categories

Specific Agents



Nature Reviews | Neuroscience



Treatment Options After Diagnosis

- Treatment is multimodal
 - Treat Underlying condition (metabolic, electrolyte, structural, etc)
 - Avoidance of precipitating factors (drugs, alcohol, sleep deprivation, stress)
 - Prophylactic therapy medications (cost risk analysis)
 - Surgery
- When to initiate prophylactic pharmacotherapy (2/3 of patients have adequate seizure control)
 - Lack family history, normal EEG, neurologic examination and MRI
 - Low incidence of seizure recurrence (15%)
 - Therapy drug toxicity may preclude necessity
 - Family history and positive EEG or MRI, abnormal neurologic exam, or seizure presenting with status epilepticus,
 - High incidence of seizure recurrence (60%)
 - Initiate therapy
 - Risk of single seizure recurrence is too high
- Pharmacoresistance therapies include, lesionectomies (focal), Vagus nerve stimulator (VNS), responsive neurostimulator (RNS), deep brain stimulation (DBS), ketogenic diet, and CBD?

Pharm Mngt.
Gen/Focal

22

First exclude: syncope, TIA, migraine, acute psychosis, and other causes of cerebral dysfunction.

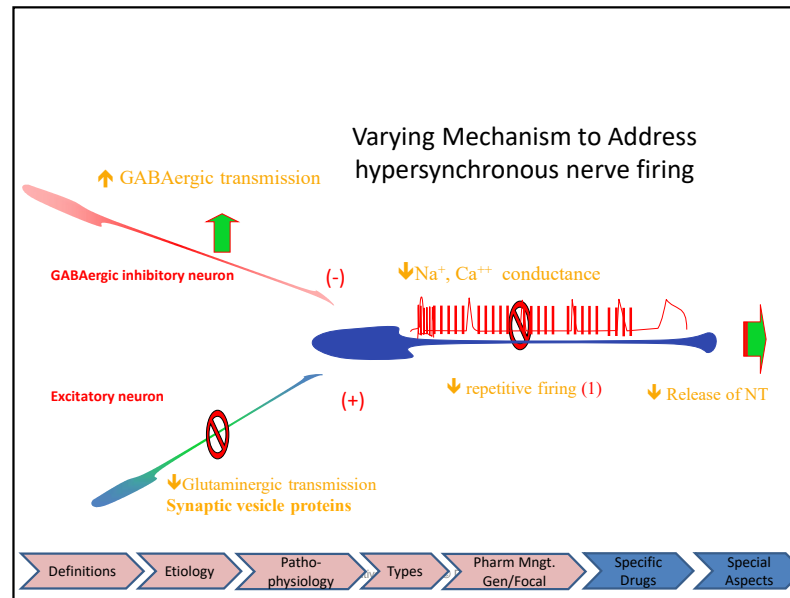
Laboratory studies follow (CBC, electrolytes, serum glucose, liver function, urinalysis, tox screen, etc.)

Note in California regarding driving:

According to CA Vehicle Code #12806c, "The department may refuse to issue to, or renew a driver's license of, any person who has a disorder characterized by lapses of consciousness or who has experienced, within the last three years, either a lapse of consciousness or an episode of marked confusion caused by any condition which may bring about recurrent lapses." Seizure-free periods of 3, 6, and 12 months are now considered when reviewing each driver's case individually. The applicant may be issued a license with restrictions such as daylight driving only, periodic medical reports, etc. (#12813).

The DMV considers the following factors when evaluating a driver with a seizure disorder:

- . Length of time driver has been seizure free
- . Type & frequency of seizures ; reliability of "warning"
- . Compliance with prescribed medical treatment
- . Scope of driver's employment
- o The driving history



These series of slides provide the background for the following testable goals:

Understand underlying molecular mechanism of GABA mimetic drugs and drugs that reduce sodium and calcium current

Know the mechanism and toxicities of all drugs described in lecture.

Note: Ideal antiseizure drug would suppress all seizures without causing any unwanted effects (G and G).

50% of the time therapy fails and unwanted effects are common

Anti seizure drugs: Effects

Table 1 | Antiepileptic drugs and their molecular targets

Drug†	Sodium channels*	Calcium channels*	GABA system*	Glutamate receptors
Predominant sodium (and calcium) channel activity				
Phenytoin	I_{NaP}			
Carbamazepine	I_{NaP}			
Oxcarbazepine	I_{NaP}			
Lamotrigine	I_{NaP}	HVA		
Zonisamide	I_{NaP}	T-type		
Mixed, complex or poorly understood actions				
Valproate	I_{NaP} ? I_{NaT} ?	T-type?	↑ GABA turnover	
Felbamate	I_{NaP}	HVA	GABA _A R	NMDA
Topiramate	I_{NaP} , I_{NaT}	HVA	GABA _A R	KA/AMPA
Ethosuximide	I_{NaT} ?	T-type		
Gabapentin		HVA ($\alpha 2\delta$)	↑ GABA turnover	
Levetiracetam‡		HVA	Reverses DMCM	
Phenobarbital		HVA	GABA _A R	AMPA

GABA-mediated mechanisms

Benzodiazepines	GABA _A R			
Vigabatrin	GABA-aminotransferase	GABAT		
Tiagabine	GABA-transporter			

A

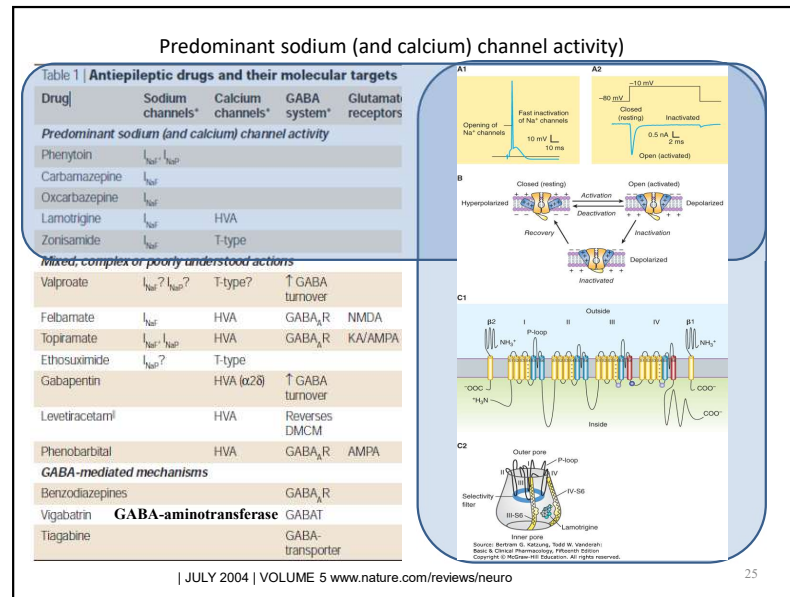
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| JULY 2004 | VOLUME 5 www.nature.com/reviews/neuro

Molecular targets for antiseizure drugs at the excitatory glutamatergic synapse (A) and the inhibitory GABAergic synapse (B). Presynaptic targets diminishing glutamate release include Na_v1.6 voltage-gated sodium channels (phenytoin, carbamazepine, lamotrigine, lacosamide, zonisamide, and oxcarbazepine); KCNQ (K_v7) voltage-gated potassium channels (retigabine [ezogabine]); and $\alpha 2\delta$ -1 protein (gabapentin and pregabalin), which interacts with NMDA receptors and voltage-gated calcium channels. Postsynaptic targets at excitatory synapses are AMPA receptors (perampanel) and KCNQ voltage-gated potassium channels (retigabine [ezogabine]). At inhibitory synapses and in astrocytes, vigabatrin inhibits GABA-transaminase (GABA-T) and tiagabine blocks GABA transporter 1 (GAT-1). Phenobarbital, primidone (via metabolism to phenobarbital), and benzodiazepines are positive allosteric modulators of synaptic GABA_A receptors; high GABA levels resulting from blockade of GABA-T may act on extrasynaptic GABA_A receptors. Citation: Chapter 24 Antiseizure Drugs, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology*, 15e; 2021. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=2988§ionid=250597700> Accessed: October 13, 2022

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24



List the mechanism and toxicities of all drugs described in lecture.

*Molecular targets. Not all molecular targets are shown; additional targets are discussed in the text. ‡Clinical efficacy of drugs on symptoms. Clinical evidence: ‘+’ indicates

controlled trials or several open-label trials and general acceptance of utility; parentheses indicate less extensive base of evidence. ‘–’ indicates evidence of lack of efficacy or

worsening. §A catastrophic epilepsy syndrome usually beginning in the first year of life in which there are typically ‘jackknife spasms’ (myoclonic seizures involving the muscles

of the neck, trunk and limbs, with nodding of the head and stiffening of the arms) and a disorganized cortical discharge termed hypsarrhythmia. || Levetiracetam binds with

high affinity to synaptic vesicle protein 2A (SV2A), a ubiquitous 90 kDa protein that is associated with synaptic vesicles and is believed to participate in the regulation of Ca²⁺-

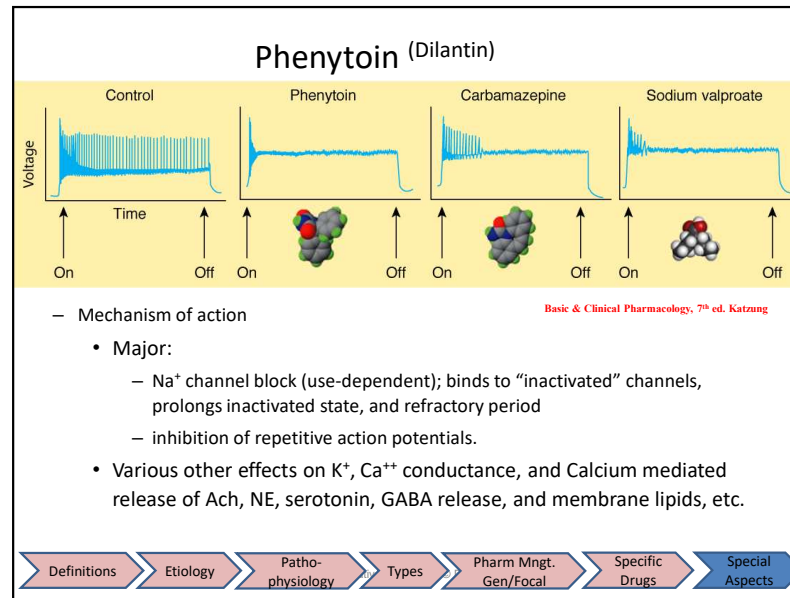
dependent neurotransmitter release; SV2A-knockout mice exhibit seizures¹³⁸. AMPA, α -amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid; GABA, γ -aminobutyric acid;

GABAAR, GABA_A receptor; GTC, generalized tonic-clonic; HVA, high voltage activated; INaF, fast sodium current; INaP, persistent sodium current; KA, kainate;

NMDA, *N*-methyl-D-aspartate. DMCM (methyl-6,7-dimethoxy-4-ethyl- β -carboline-3-carboxylate) is a negative allosteric modulator of GABA_A receptors. Adrenocorticotrophic

hormone and prednisolone are recognized treatments for infantile spasms. Lamotrigine is also effective in myoclonic astatic epilepsy, but causes worsening of other forms of

myoclonic epilepsy. Gabapentin has been associated with focal myoclonus.



Testable objective: Know everything about phenytoin

Phenytoin (Dilantin) cont'd

- Pharmacokinetics:
 - Absorption
 - from intestine: complete, 3-12 h depending on formulation. Usually dissolved in propylene glycol which can be cardiotoxic
 - from intramuscular: unpredictable, precipitates, therefore new soluble form: **fosphenytoin** (*Cerebyx*) (converts to phenytoin in plasma, also, does not need PEG to be soluble)
 - Binding
 - to plasma proteins ([CSF] = free drug in plasma)
 - endoplasmic reticulum (brain, liver, other tissues)



Testable objective: Know everything about phenytoin

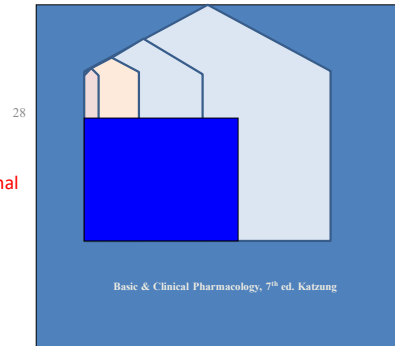
Phenytoin (Dilantin) cont'd

- Phenytoin Pharmacokinetics (Cont.):

- Metabolism: liver (CYP2C9/10 and CYP2C19)
 - para-hydroxylation, then glucuronidation
 - metabolites are inactive, excreted in urine

- DOSE DEPENDENT (saturable)

- at low doses: 1st order (proportional to dose)
- at therapeutic doses: close to liver metabolizing capacity saturation -- small $\uparrow$ in dose = large $\uparrow$ in [plasma]

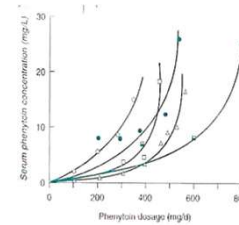


Testable objective: Know everything about phenytoin

Note: 95% metabolized by CYP2C9/10 which can be saturated

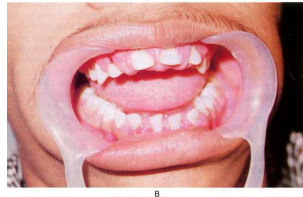
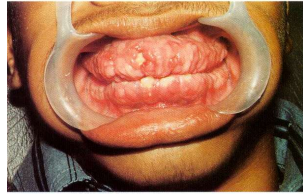
Phenytoin (Cont.)

- Therapeutic Levels and Dosage
 - Therapeutic plasma level 10-20 ug/mL
 - » toxic level >20 ug/ml
 - In *status epilepticus*: intravenous loading
 - Oral dosage:
 - Typical starting dose 300 mg/d; frequently insufficient to reach ther. plasma level.
 - Even small increments may result in toxic levels, therefore 25-30 mg steps are advised (5mg in child)
 - Preparations:
 - Phenytoin: Oral, oral-extended (1x/d), and parenteral
 - Fosphenytoin inj. (more soluble, replacing phenytoin)



Testable objective: Know everything about phenytoin

Phenytoin: Toxicity



Gingival Hyperplasia Induced by Phenytoin

- **common for the group (dose dependent)**
 - Nystagmus, early
 - Ataxia, diplopia (adjust dose!)
 - Sedation at high plasma level
 - Gingival hyperplasia, hirsutism (very common-- 20%)
 - Long term:
 - Peripheral neuropathy, diminished reflexes (30%)
 - ↑ Vitamin D liver metabolism (enzyme induction) → ↑ Osteomalacia, ↑ fracture rate (1.2 to 2.4 times in elderly)
 - ↓ folic acid absorption and plasma levels (pH alterations in gut?) → Megaloblastic anemia (rare, <1%), supplements help
 - Idiosyncratic: rare
 - rare, skin rash (2-5%), fever (with agranulocytosis)
 - lymphadenopathy, Hodgkin's ?

Definitions

Etiology

Patho-physiology

Types

Pharm Mngt.
Gen/Focal

Specific
Drugs

Special
Aspects

Testable objective: Know everything about phenytoin

Phenytoin and Other Hydantoins

Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Hydantoin	*Phenytoin (Dilantin)	dec. repetitive neuronal firing	state dependent Na ⁺ channel blockade (inactive state). inhibition of repetitive AP; prolongs inactivated state. Ca ⁺⁺ mediated neurotransmitter release.	Tonic-clonic Focal-onset	dose dependent + saturable; pseudozero- order kinetics (1st order at low dose, at therapeutic doses: close to liver metabolizing capacity saturation - small inc. in dose = large inc in [plasma])	Induces CYP 3A4 enzymes (many drug interactions, .eg. Increases metabolism of oral contraceptives). Serum protein binding displaced by other drugs (ex: sulfonamides) leading to inc [free]. Level increased by isoniazid, sulfonamides, fluoxetine Binds thyroid-binding globulin	Nystagmus, ataxia, diplopia, sedation neuropathy, gingival hyperplasia, osteomalacia (due to inc. Vit. D liver metabolism), hirsutism, skin rash, megaloblastic anemia (rare, due to low folate), agranulocytosis
	Fosphenytoin (Cerebyx)				IV, IM: more soluble IV form, PEG not needed to dissolve. Converted in body to phenytoin		
	Mephenytoin (Mebaral)			similar to phenytoin; less efficacious	similar to phenytoin, nirvanol is an active metabolite	Level decreased by en- zyme-inducing drugs. Altered folate metabolism	Dermatitis, agranulocytosis, hepatitis more frequent than phenytoin
	Etholoin (Peganone)			similar to phenytoin	similar to phenytoin		less severe than phenytoin

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Protein Binding

displaced by other drugs (e.g. sulfonamides), leading to □ [free]

hypoalbuminemia-raising dose to maintain total drug concentration may cause intoxication

interferes with thyroid function tests because phenytoin binds to thyroid-binding globulin (use TSH assay)

Metabolism

↑ microsomal enzyme induction

effects metabolism of other drugs (e.g. oral contraceptives)

↑ steroid and steroid vitamins (D)

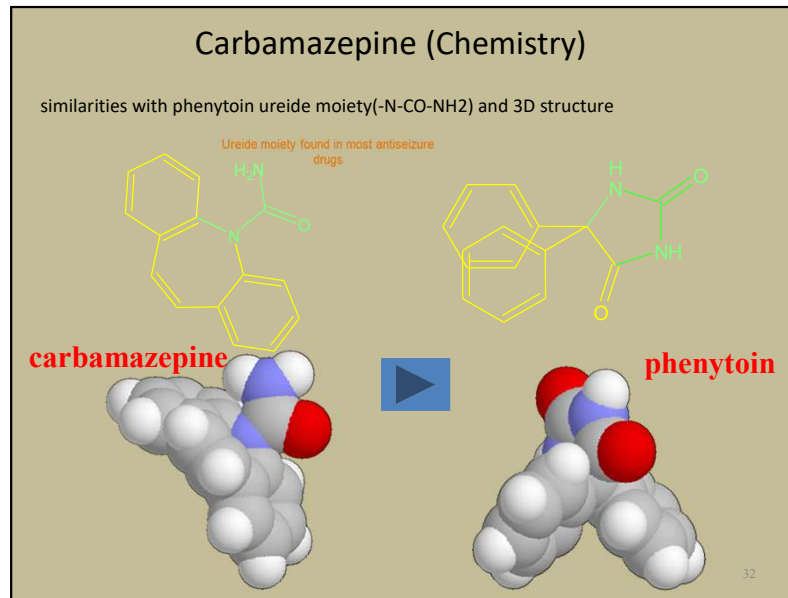
Hepatic microsomal enzyme stimulation (by phenobarbital,

carbamazepine)

↑ metabolism, ↓ [steady-state] of phenytoin.

Isoniazid inhibits metabolism of phenytoin

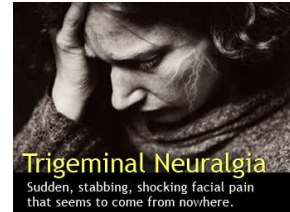
Folic acid may enhance phenytoin metabolism



This slide illustrates the structural basis to why carbamazepine and phenytoin are used interchangeable for seizure management.

Carbamazepine (Cont.) ~

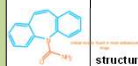
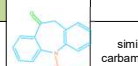
- Mechanism of action: similar to phenytoin
 - blocks Na⁺ channels, inhibits repetitive firing in cells
 - decreases synaptic transmission
- Use:
 - Current, 1st line drug for Focal (partial) seizures (with others)
 - Trigeminal neuralgia
 - Alternative to lithium to treat Bipolar disorders



Know the mechanism and toxicities of all drugs described in lecture.

Note Carbamazepine also has antidiuretic properties

Carbamazepine and Oxcarbazepine

Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Tricyclic	*Carbamazepine (Tegretol)	 dec. repetitive neuronal firing—↑ Na channel inactivation	structure similar to phenytoin, also inhibition of catecholamine uptake	Tonic-clonic Focal-onset bipolar disorder, trigeminal neuralgia (DOC).	-	CYP inducer Level decreased by enzyme-inducing drugs Level increased by erythromycin, propoxyphene, isoniazid, cimetidine, fluoxetine	Vertigo, drowsiness, diplopia, ataxia; GI upset; hepatotoxicity, hyponatremia, Aplastic anemia, agranulocytosis, reversible leukopenia; hypersensitivity for high risk Asians w/ HLA-B*57:01 allele
	Oxcarbazepine		similar to carbamazepine	similar to carbamazepine (less potent)	active metabolite – S(+) and R(-) licarbazepine	weaker induction of hepatic enzymes - fewer Rx interactions	similar to carbamazepine but less toxicity?
	Eslicarbazepine		Similar to oxcarbazepine		Active metabolite – S(+) licarbazepine		

Drug-drug interaction: More enzymes = ↑ metabolism = ↓ effect

- General Inducers: Barbiturates, Phenytoin, Rifampin, Griseofulvin, Carbamazepine
- Barb takes Phen-phen & Refuses Greasy Carbs

Know the mechanism and toxicities of all drugs described in lecture.

GI upset includes nausea, vomiting

Hypersensitivity reactions include dermatitis, eosinophilia, lymphadenopathy, splenomegaly, Stevens Johnson syndrome in Asians with specific HLA allele

Late complications is water retention, with decreased osmolality and concentration of sodium in plasma

Transient elevation in hepatic enzymes (5-10%) of patients.

Transient mild leukopenia (10%), and thrombocytopenia

Aplastic anemia (1/200,000) patients

Oxcarbazepine--Like carbamazepine, but less toxic

Glucuronide--Active metabolite (10-hydroxycarbamazepine)

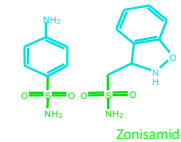
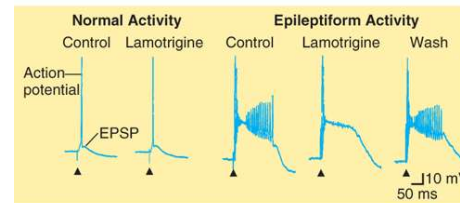
Fewer hypersensitivity reactions

Weaker induction of hepatic enzymes - fewer drug interactions

Lamotrigine and Zonisamide

Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Miscellaneous	Lamotrigine (Lamictal)	dec. repetitive neuronal firing	Blocks Na ⁺ channels & CC channel activity	1st line for all seizures EXCEPT absence type seizure (2nd line) Focal-onset, Tonic-clonic, Atypical absence, Myoclonic, Lennox-Gastaut, syndrome Also for Bipolar Depression	rapidly absorbed, glucuronidated, 55% protein bound	valproate doubles half life, decreased by CYP inducing drugs and oral contraceptives	Dizziness, Diplopia, Sedation, Ataxia, Headache skin rash: 1-2% life threatening in children-- Stevens-Johnson syndrome
Sulfonamide	Zonisamide	Dec sodium and calcium channel conductance	unknown	Focal-onset Tonic-clonic	oral, good bioavailability	little effect to other antiseizure drugs. Level decreased by enzyme-inducing drugs	drowsiness, cognitive impairment, headache, psychosis anorexia, renal stones, hypohidrosis

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Selective effect of a clinically relevant concentration of lamotrigine (50 μ M) on action potentials and epileptic-like discharges in rat hippocampal neurons as assessed with intracellular recording. In normal recording conditions, lamotrigine has no effect on action potentials or on the evoked excitatory postsynaptic potentials (EPSPs) that elicit the action potential. In epileptic-like conditions (low magnesium), activation elicits initial spikes followed by repetitive epileptiform spike firing (afterdischarge). Lamotrigine inhibits the pathologic discharge but not the initial spikes. EPSPs were elicited by stimulation of the Schaffer collateral/commissural fibers (triangles). (B) Voltage and use dependence of block of human Na_v1.2 voltage-activated sodium channels. Sodium currents elicited by depolarization from a holding potential of -90 mV (where there is little inactivation) are minimally affected by 100 μ M of lamotrigine, whereas there is strong block of current elicited from -60 mV (where there is more substantial inactivation). Trains of 0.7-millisecond (ms) duration pulses from -90 mV (minimal inactivation) are minimally blocked in a use-dependent fashion by 100 μ M of lamotrigine, whereas 20-ms pulses (marked inactivation) show substantial use dependence. (Adapted with permission from Xie X, Hagan RM: Cellular and molecular actions of lamotrigine: Possible mechanisms of efficacy in bipolar disorder, *Neuropsychobiology* 1998 Oct;38(3):119-130.) Citation: Chapter 24 Antiseizure Drugs, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology*, 15e; 2021. Available at:

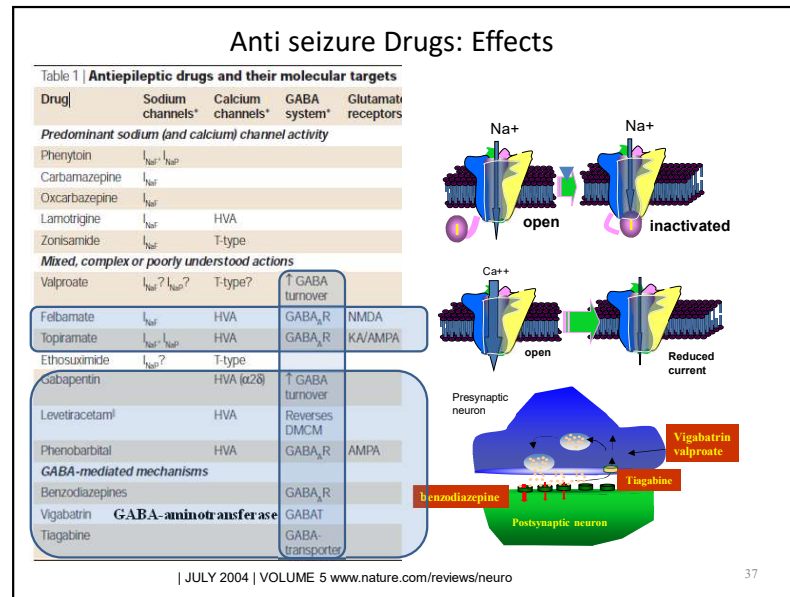
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Sulfonamide	Zonisamide	Dec sodium and calcium channel conductance	unknown	Focal-onset Tonic-clonic	oral, good bioavailability	little effect to other antiseizure drugs. Level decreased by enzyme-inducing drugs	drowsiness, cognitive impairment, headache, psychosis anorexia, renal stones, hypohidrosis





List the mechanism and toxicities of all drugs described in lecture.

HVA is High voltage activated calcium channels

*Molecular targets. Not all molecular targets are shown; additional targets are discussed in the text. ‡Clinical efficacy of drugs on symptoms. Clinical evidence: ‘+’ indicates

controlled trials or several open-label trials and general acceptance of utility; parentheses indicate less extensive base of evidence. ‘–’ indicates evidence of lack of efficacy or

worsening. §A catastrophic epilepsy syndrome usually beginning in the first year of life in which there are typically ‘jackknife spasms’ (myoclonic seizures involving the muscles

of the neck, trunk and limbs, with nodding of the head and stiffening of the arms) and a disorganized cortical discharge termed hypsarrhythmia. || Levetiracetam binds with

high affinity to synaptic vesicle protein 2A (SV2A), a ubiquitous 90 kDa protein that is associated with synaptic vesicles and is believed to participate in the regulation of Ca²⁺-

dependent neurotransmitter release; SV2A-knockout mice exhibit seizures¹³⁸. AMPA, α-amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid; GABA, γ-aminobutyric acid;

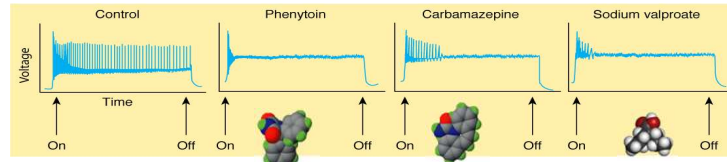
GABAAR, GABA_A receptor; GTC, generalized tonic-clonic; HVA, high voltage activated; INaF, fast sodium current; INaP, persistent sodium current; KA, kainate;

NMDA, N-methyl-D-aspartate. DMCM (methyl-6,7-dimethoxy-4-ethyl-β-carboline-3-carboxylate) is a negative allosteric modulator of GABA_A receptors. Adrenocorticotrophic

hormone and prednisolone are recognized treatments for infantile spasms. Lamotrigine is also effective in myoclonic astatic epilepsy, but causes worsening of other forms of

myoclonic epilepsy. Gabapentin has been associated with focal myoclonus.

Valproic acid (Depakene) and Na valproate



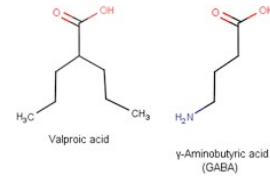
Discovered when used as **solvent** for other anti-epileptics

- Chemistry:

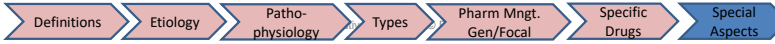
- Effective chain-length 5-8 C
- Branching and unsaturation $\uparrow$ lipophilicity, and duration

- MOA

- Direct ion channel effects?
 - 1 $\uparrow$ VG-Na⁺ channel inactivation)
 - 2 Limits activity of VGCC-t-type
 - 3 $\uparrow$ Membrane potassium conductance (high levels)
- $\uparrow$ GABA levels in brain
 - 1 $\uparrow$ synthesis via increase in glutamic acid decarboxylase
 - 2 $\downarrow$ breakdown via inhibition of GABA transaminase
 - 3 $\downarrow$ reuptake
 - 4 $\uparrow$ GABA turnover
- Altered gene expression via Chromatin remodeling inhibition of histone deacetylases?



Neuroscience & Medicine
Vol. 3 No. 1(2012), Article ID: 17746, 14 pages
DOI:10.4236/nmm.2012.31016



Mixed Mechanism/poorly understood MOA							
Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Compounds with GABA like structure	Valproic Acid (Depakene, Depakote)	dec. repetitive firing, dec. Ca ⁺⁺ conductance, inc. GABAergic transmission	Inhibits Na ⁺ currents, increased GABA levels, inc. membrane K ⁺ conductance and blockade of T-type Ca ²⁺ channels in thalamus.	Broad spectrum: Tonic-clonic, Absence (ethosuximide preferred), Atypical absence Myoclonic, Focal-onset Atonic; other uses: bipolar disorder, migraines	good oral absorption, highly ionized and protein bound, dose dependent clearance., Various compositions Na, Mg salts, syrup, free acid; "coordination compound" favored by patients because of slow absorption	levels decrease by enzyme-inducing agents, displaces phenytoin from protein binding	Ataxia, Sedation, Tremor, Nausea, vomiting and GI complaints, Hepatotoxicity (sometimes fatal, monitor liver fn), weight gain , transient alopecia, Teratogenic: spina bifida
Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Miscellaneous	Topiramate (Topiramate)		probably effects Na channels, Kainate and AMPA receptor inhibition, potentiates GABA inhibitory effects	Focal-onset Tonic-clonic Lennox-Gastaut syndrome Alternative for Partial type seizures; also used for preventative therapy for migraine headaches	administer slowly and inc. dose	birth control pills are less effective. Level decreased by enzyme-inducing drugs	Significant toxicity issue: dose-related (GIVE DRUG SLOWLY) , in first 4 wks, significant psychomotor and cognitive symptoms (somnolence, fatigue, dizziness, confusion), also inc. IOP, caution w/ nephrolithiasis
	Trimethadione (Tridione)	dec. Ca ⁺⁺ conductance, raises threshold for firing of repetitive thalamic firing	similar to ethosuximide effects on gCa ⁺⁺	third line for absence seizures behind succinimides	good oral absorption, not protein bound, liver metabolized	minor	sedation, hemeralopia, mild metabolic acidosis, idiosyncratic derm reactions, reversible nephrotic syndrome
	Levetiracetam (Keppra)	dec. glutamatergic transmission		adjunctive therapy for partial seizures	mainly excreted in urine	minimal	somnolence, asthenia and dizziness. Rare: anemia, leukopenia
	Zonisamide	sodium and calcium channel conductance	unknown	Focal-onset Tonic-clonic	oral, good bioavailability	little effect to other antiseizure drugs, Level decreased by enzyme-inducing drugs	drowsiness, cognitive impairment, headache, psychosis, anorexia, renal stones, hypohidrosis
	Bromide	unclear	unknown	management of epilepsy patients with porphyria			skin rash, sedation, and behavioral changes

Valproic acid

Drug interactions:

displaces phenytoin from protein;

inhibits the metabolism of phenytoin, carbamazepine; phenobarbital (the latter may cause stupor or coma)

Toxicity

Gastrointestinal (common, dose-related): nausea, vomiting, abdominal pain, heartburn - gradual start, temporary reduction helps

Sedation: rare with valproate, but strong when valproate is added to phenobarbital

Idiosyncratic : common elevation in hepatic enzymes, rare but severe hepatotoxicity (>50 fatal in US) - check liver function.

Fetal abnormalities reported - (? in pregnancy)

Use:

Very effective for absence seizures, (but because of its hepatotoxicity, ethosuximide is preferred)

First line for monoclonic seizures

grand mal

Bipolar disorders and management of migraines

Pharmacokinetics

Well absorbed (>80%), taking with meals delays absorption, □ toxicity, Fully ionized (anion is active. 90 % Bound to plasma protein,

half-life 9-18 h; dose-dependent

Various compositions

Na, Mg salts, syrup, free acid;

“coordination compound”- favored by patients because of slow absorption

Topiramate (Topamax)

unique, substituted monosaccharide

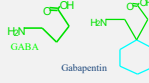
Key Issues

Also used for preventative therapy for migraine headaches

Significant toxicity issue,

- Toxicity: dose-related (give drug slowly), in first 4 weeks, Significant psychomotor and cognitive symptoms (somnolence, fatigue dizziness, confusion), also increased IOP, caution with nephrolithiasis

Mixed Mechanism/poorly understood MOA

Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Compounds with GABA like structure	Felbamate (Felbatol)	Inc. GABAergic Transmission, dec. Glutamatergic transmission		alternative for all except Absence type seizure		inc. plasma phenytoin and valproic acid but dec. carbamazepine levels	Insomnia, Dizziness, Sedation, Headache significant toxicity issue: aplastic anemia and hepatitis
	Vigabatrin	Inc. GABAergic Transmission	inhibits GABA transaminase (results in more GABA release)	Partial seizures and West's syndrome, infantile spasm (with corticotropin and prednisone)	activity is not correlated with plasma half life, eliminated via kidneys		drowsiness, dizziness and wt. Gain, less common: psychosis and confusion and visual field defects, animal studies: intramyelitic edema
	Tiagabine (Gabitril)	Inc. GABAergic Transmission	inhibits GABA uptake (GAT-1 transporter isotype)	adjunctive therapy for partial seizures	oral absorption, good bioavailability	does not effect levels of other seizure drugs. Level decreased by enzyme-inducing drugs	Nervousness, dizziness, tremor, difficulty concentrating and depression. Discontinue if psychosis develops. Rash common, GI irritation
	Gabapentin (Neurotonin)	Inc. GABAergic transmission, dec. Ca++ conductance	Inc. GABA synthesis(glutamic acid decarboxylase) Inhibits HVA CC	Adjunct therapy for partial seizures, effective for neuropathic pain (postherpetic neuralgia)	not metabolized renal excretion, short half life	Other CNS depressants	CNS depression in any drug that increases GABA effects (sedation, dizziness, ataxia, fatigue), GI irritation, weight gain, edema
 Designed to bind to GABA channels but does not. Main use is pain modulation							

2013;119(5):1215-1221.
doi:10.1097/ALN.0b013e3182a9a896

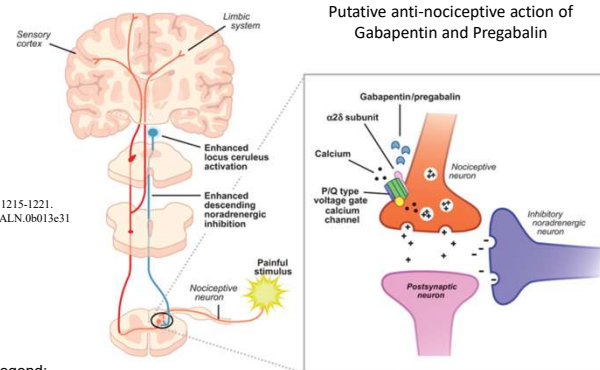


Figure Legend:

Paraphrased from original: Gabapentinoids (Gabapentin and pregabalin) were designed to function like GABA. However the mechanism of action appears to involve inhibiting high velocity calcium channels (HVA) the voltage-gated P/Q-type type calcium channels at the α -2 δ subunit. The reduction in calcium entry into pre-synaptic dorsal horn neuron reduces docking and fusion of synaptic vesicles and the release of neurotransmitters resulting in a reduction of activation of post-synaptic ascending fibers. In addition, gabapentin may exert an analgesic effect by activating descending inhibitory noradrenergic pathways that regulate neurotransmission of pain signals in the dorsal horn of the spinal cord.

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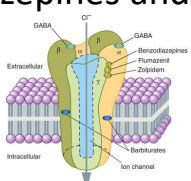
Benzodiazepines and Barbiturates



Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Barbiturate	*Phenobarbital	inc. GABAergic transmission	inc. GABA transmission, effects Na channel opening	2nd line for Primary Generalized Tonic-clonic seizure; Partial seizure; for children; not for long term use	see sedative chart	see sedative chart	relatively safe but marked sedation
	Mephobarbital						
	Primidone		Active metabolite = phenobarbital				similar to phenobarbital

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Primidone itself is similar to phenytoin in antiepileptic effects. Metabolism produces active metabolites (phenobarbital and PEMA). These products accumulate slowly (steady state in 20 days!)

Benzodiazepines and Barbiturates							
							
Source: Bertram G, Katzung, Anthony J, Trevor, Basic & Clinical Pharmacology, 12th Ed.							
Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Benzodiazepines	Diazepam (Valium), Lorazepam (Ativan)			Status Epilepticus	Lorazepam more effective and durable than diazepam. Chronic oral use of diazepam ineffective because of tolerance		
	Clonazepam (Klonopin), Clorazepate (Tranxene-SD),	Inc. GABAergic Transmission	allosteric activator of GABA mediated chloride conductance	long term seizure management (tolerance and sedation limits chronic use); Infantile spasms (if seizures recur) with ACTH and prednisone; Clonazepam - for Absence type seizure, also works for myoclonic s.; Clorazepate as adjunct Rx for complex partial seizure.	Both have strong sedative effect, start with low doses		see sedative lecture

Primidone itself is similar to phenytoin in antiepileptic effects. Metabolism produces active metabolites (phenobarbital and PEMA). These products accumulate slowly (steady state in 20 days!)

Tolerance and Sedation limits chronic use

Clorazepate (Tranxene-SD) : adjunct drug for complex Focal (partial) s. Drowsiness, lethargy common, raise doses slowly.

Clonazepam (Klonopin) : Long acting.

One of the best for absence s., works for myoclonic s. Prominent sedative effect, start with low doses, raise over weeks.

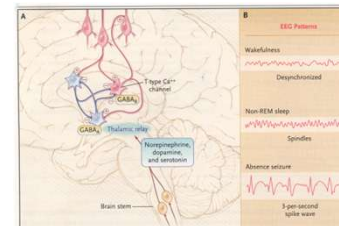
Lorazepam (Ativan) I.V. for status epilepticus; can be more effective and durable than diazepam.

Diazepam (Valium): stops seizures, I.V. or rectally, for status epilepticus. Chronic oral use ineffective because of tolerance.

Table 1 Antiepileptic drugs and their molecular targets				
Drug	Sodium channels*	Calcium channels*	GABA system*	Glutamate receptors
Predominant sodium (and calcium) channel activity				
Phenytoin	I_{NaP}			
Carbamazepine	I_{NaP}			
Oxcarbazepine	I_{NaP}			
Lamotrigine	I_{NaP}	HVA		
Zonisamide	I_{NaP}	T-type		
Mixed, complex or poorly understood actions				
Valproate	I_{NaP} ? I_{NaP} ?	T-type?	↑ GABA turnover	
Felbamate	I_{NaP}	HVA	GABA _A R	NMDA
Topiramate	I_{NaP}	HVA	GABA _A R	KA/AMPA
Ethosuximide	I_{NaP} ?	T-type		
Gabapentin		HVA (α2δ)	↑ GABA turnover	
Levetiracetam		HVA	Reverses DMCM	
Phenobarbital		HVA	GABA _A R	AMPA
GABA-mediated mechanisms				
Benzodiazepines			GABA _A R	
Vigabatrin	GABA-aminotransferase		GABAT	
Tiagabine			GABA-transporter	

| JULY 2004 | VOLUME 5 www.nature.com/reviews/neuro

Ethosuximide



List the mechanism and toxicities of all drugs described in lecture.

HVA is High voltage activated calcium channels

*Molecular targets. Not all molecular targets are shown; additional targets are discussed in the text. ‡Clinical efficacy of drugs on symptoms. Clinical evidence: ‘+’ indicates

controlled trials or several open-label trials and general acceptance of utility; parentheses indicate less extensive base of evidence. ‘–’ indicates evidence of lack of efficacy or

worsening. §A catastrophic epilepsy syndrome usually beginning in the first year of life in which there are typically ‘jackknife spasms’ (myoclonic seizures involving the muscles

of the neck, trunk and limbs, with nodding of the head and stiffening of the arms) and a disorganized cortical discharge termed hypsarrhythmia. || Levetiracetam binds with

high affinity to synaptic vesicle protein 2A (SV2A), a ubiquitous 90 kDa protein that is associated with synaptic vesicles and is believed to participate in the regulation of Ca²⁺-

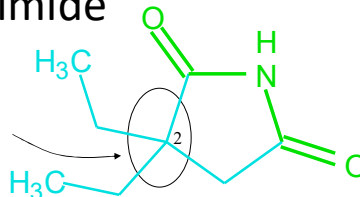
dependent neurotransmitter release; SV2A-knockout mice exhibit seizures¹³⁸. AMPA, α-amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid; GABA, γ-aminobutyric acid;

GABAAR, GABA_A receptor; GTC, generalized tonic-clonic; HVA, high voltage activated; INaF, fast sodium current; INaP, persistent sodium current; KA, kainate;

NMDA, N-methyl-D-aspartate. DMCM (methyl-6,7-dimethoxy-4-ethyl-β-carboline-3-carboxylate) is a negative allosteric modulator of GABA_A receptors. Adrenocorticotrophic

hormone and prednisolone are recognized treatments for infantile spasms. Lamotrigine is also effective in myoclonic astatic epilepsy, but causes worsening of other forms of

myoclonic epilepsy. Gabapentin has been associated with focal myoclonus.

Ethosuximide -substituted succinimide + uriede (if phenyl: phensuximide, methsux.) 							
Class	Drug	Effects	MOA	Clinical Use	P-kinetics	Drug Interactions	Toxicity
Succinimides	Ethosuximide (Zarontin)	dec. Ca ⁺⁺ conductance	inhibits gCa ⁺⁺ , Na ⁺ /K ⁺ atpase and inhibits GABA aminotransferase, reduces thalamic firing through blockade of T-type Ca ²⁺ channels	first line for absence (petit mal); only use for ethosuximide	completely absorbed peaks after 3-7 hours, no protein binding; plasma conc. = CSF conc; completely metabolized by hydroxylation, conjugation and renal excretion; slow clearance, long half life (18-70h)	level decreased by enzyme-inducing drugs	gastric distress, transient lethargy, rare skin rash and more marrow suppression (periodic monitoring necessary)
Definitions Etiology Patho-physiology Types Pharm Mngt. Gen/Focal Specific Drugs Special Aspects							

Ethosuximide (Cont.)

Use: Absence seizures

documented by long-term EEG (3/s spike-waves)

Pharmacokinetics:

Completely absorbed peaks after 3-7 h.

No protein binding; plasma conc. = CSF conc.

Completely metabolized by hydroxylation, conjugation, renal excretion

Slow clearance. long half-life (18 to 70 h)

Dosage: plasma level linear with dose.

Once a day (or twice because of gastroint. effects)

Toxicity

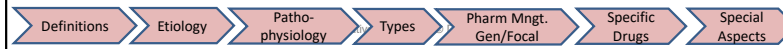
gastric distress, transient lethargy, rare skin rash and more marrow suppression (periodic monitoring necessary)

Drug Choice

Confidence level

Efficacy vs Toxicity

Pharmacokinetics (e.g. once a day)



Harrison's Principles of Internal Medicine, 19e, 2015 > Seizures and Epilepsy
Dennis Kasper, Anthony Fauci, Stephen Hauser, Dan Longo, J. Larry Jameson, Joseph Loscalzo+
TABLE 445-8 Selection of Antiepileptic Drugs

Drug Choice	Primary Generalized-Onset Tonic-clonic	Focal (partial)	Typical Absence	Atypical Absence, Myoclonic, Atonic
First-Line	Lamotrigine Valproic acid	Lamotrigine Carbamazepine Oxcarbazepine Esclicarbazepine Phenytoin Levetiracetam	Valproic acid Ethosuximide Lamotrigine	Valproic acid Lamotrigine Topiramate
Alternatives	Zonisamide ^a Phenytoin Carbamazepine Oxcarbazepine Topiramate Phenobarbital Primidone Felbamate	Topiramate Zonisamide ^a Valproic Acid Tiagabine ^a Gabapentin ^a Locosamide ^a Phenobarbital Primidone Felbamate Perampanel	Lamotrigine (1st or alternative) Clonazepam Zonisamide Levetiracetam	Clonazepam Felbamate Clobazam Rufinamide

Focal (partial) seizures includes simple Focal (partial), complex Focal (partial), and secondarily generalized seizures. a= adjunctive agents

Know which drugs are used for the specific seizure types

Note e.g. of drug choice described below:

Drug Selection for Focal (partial) Seizures:

Efficacy: Carbamazepine, phenytoin, lamotrigine, and topiramate have similar efficacy

Pharmacokinetics: phenytoin is once a day vs some others which may be three times. However, saturable kinetics leads to toxicity.

Toxicity: Phenytoin many with chronic use, while carbamazepine has serious hematopoietic toxicities. Lamotrigine has less toxicities but can cause Stevens-Johnson syndrome (must monitor with initial doses). Topiramate is associated with significant psychomotor slowing and other cognitive issues. Phenobarbital is second line because of sedation and cognitive issues. Other toxicity issues to consider--decreased libido and impotence (carbamazepine 13%, phenobarbital 16%, phenytoin 11% and primidone 22%)

Drug selection for generalized seizures:

Phenytoin and carbamazepine can worsen some types of generalized seizures and are therefore second line.

Absence Seizures: ethosuximide and valproate are equally effective (50-75% patients benefit)

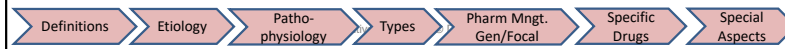
Special Aspects

Women who are pregnant or on contraceptives

Withdrawal

Overdose

When to discontinue?



Toxicology: Special aspects (Cont.) ~

- Pregnancy
 - Seizure frequency
 - Unchanged in 50% and increased in 30%
 - Teratogenicity
 - Chronic treatment of millions of patients, but few untreated epileptic “controls”
 - About two-fold increase of congenital malformations or 5-6% v 2-3%
 - Fetal hydantoin syndrome - questionable
 - **Valproate**: overall increase 7-20%, spina bifida in 1-2%
 - Current recommendation:
 - Maintain therapy, monotherapy if possible at lowest effective dose.
 - Folic acid supplementation
- Contraception
 - Carbamazepine, phenytoin, phenobarbital, and topiramate are CYP inducers that increase the metabolism of oral contraceptives.



Toxicology: Special aspects (Cont.) ~

- Withdrawal
 - Increased seizure frequency and severity with all
 - Barbiturates and benzodiazepines are most difficult to stop: needs weeks or months
- Overdose
 - Rarely lethal
 - Respiratory depression, potentiated by alcohol
 - Treatment: supportive, no stimulants are used.
- When to discontinue
 - 70% of children and 60% of adults who are under complete control can eventually discontinue if:
 - 1-5 years seizure free
 - Single seizure type
 - Normal neurologic exam and cognition
 - Normal EEG
 - Gradually taper dose over 2-3 months to avoid withdrawal and chance of recurrence and avoid dangerous situations



Drug List

- Phenytoin, fosphenytoin, mephenytoin, ethotoin
- Carbamazepine, oxcarbazepine
- Phenobarbital, mephobarbital
- Primidone
- Valproic Acid
- Ethosuximide
- Diazepam, clonazepam, clorazepate, lorazepam,
- Lamotrigine
- Felbamate
- Topiramate
- Gabapentin
- Tiagabine
- Trimethadione
- Zonisamide

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Study Goals

- Describe the pathophysiology of Focal (partial) and generalized seizure and how this relates to GABAergic and Glutaminergic neuronal activity
- Be able to identify classic symptoms of each type of seizure discussed.
- Recognize the EKG patterns, especially with petit mal (Absence seizures)
- List drugs associated with decreasing seizure threshold
- Define excitotoxicity and relate it to status epilepticus
- Understand the molecular role of GABA mimetic drugs and drugs that decrease sodium and calcium current in reducing seizure activity.
- Distinguish the role of thalamic calcium signaling and the subsequent treatment management of in petit mal seizures
- Categorize which drugs are used for the specific seizure types
- Understand the cost risk benefit ratio involved in the treatments for petit mal vs grand mal type seizures
- Know everything about phenytoin
- List the mechanism and toxicities of all drugs described in lecture.
- Drug List: Phenytoin, fosphenytoin, mephenytoin, ethotoin, Carbamazepine, oxcarbazepine, Phenobarbital, mephobarbital, Primidone, Valproic Acid, Ethosuximide, Diazepam, clonazepam, clorazepate, lorazepam, Lamotrigine, Felbamate, Topiramate, Gabapentin, Tiagabine, Trimethadione, Zonisamide